Shahjalal Univer... ...ce and Technology, Sylhet
3rd Year 2nd Semester Final Examination 2023 (Session: 2020-2021)
Subject : Mathematics
Course No.: MAT 327; Course Title: Numerical Analysis
Marks: 60 Credits: 3.0 Time: 3 Hours

Instructions : Answer 05 (Five) questions taking at least 02 (two) from each group.

Group - A

Question 1. Marks: 2+5+5=12

a) What do you mean by numerical methods ? Define absolute, relative and percentage errors in numerical methods.

b) Discuss the method of bisection for finding a real root of the equation $f(x)=0$. Find the real root of the equation $x^3-2x-5=0$, correct upto 2 decimal places.

c) What is the method of fixed point iteration? How can you confirm the convergence of this method? Perform 3 iterations for the equation $2x=\cos x+3$ with the initial approximation $x_0=\pi/2$ by this method.

Question 2. Marks: 6+6=12

a) Derive Newton-Raphson formula to solve a nonlinear equation $f(x)=0$. Give the geometrical interpretation of the Newton-Raphson method.

b) Find a root, correct to three decimal places, of the equation $\cos x - xe^x = 0$ in the interval $(0, 1)$ using Regula-Falsi method.

Question 3. Marks: 6+6=12

a) Define interpolation. Derive the Newton's formula for backward interpolation.

b) Derive the expression for the error in polynomial interpolation. Then find the error for Newton's forward difference interpolation formula.

Question 4. Marks: 6+6=12

a) Construct a central difference table for the data points (x_i, y_i); $i=0,1,2,3,4$ and hence write the Stirling formula. Use this scheme to find y for $x=35$ from the data sets:

x :	20	30	40	50
y :	512	439	346	243

b) What is divided- differences? Consider the data points:

x :	-1	0	1	2
$f(x)$:	3	-4	5	-6

Use Newton's interpolatory divided- difference formula to construct a cubic interpolatory polynomial and hence evaluate $f(1.5)$.

Group - B

Question 5. Marks: 12

Define spline functions. What do you mean by quadratic spline interpolation? Given the data points $(0,0)$, $(1,1)$ $(2,4)$, fit a quadratic spline to the function P defined by this set of points and use it to approximate $P(1.5)$.

Question 6. Marks: 12

Derive the three-point endpoint and mid point formulas to evaluate $f'(x)$ and $f''(x)$ from a given set of values. Use all the appropriate three-point formulas to approximate $f'(1.9)$, $f'(2)$, $f''(1.9)$ and $f''(2)$:

x :	1.8	1.9	2.0	2.1	2.2
$f(x)$:	10.889	12.703	14.778	17.149	19.855

Question 7. Marks: 12

Estimate the value of the integral $\int_4^{5.2} \ln x\,dx$ by (i) Trapezoidal rule, (ii) Simpson's $\frac{1}{3}$ rule, (iii) Simpson's $\frac{3}{8}$ rule and (iv) Weddle's rule with $h=0.2$. After finding the true value of the integral, compare the errors in the four cases.

Question 8. Marks: 6+6=12

a) Find the solution of the following system using Gaussian elimination with partial pivoting :

$$\begin{aligned} 20x_1 + 15x_2 + 10x_3 &= 45 \\ -3x_1 - 2.249x_2 + 7x_3 &= 1.751 \\ 5x_1 + x_2 + 3x_3 &= 9 \end{aligned}$$

b) Solve the following tri-diagonal system:

$$\begin{aligned} 2x_1 - x_2 &= 1 \\ -x_1 + 2x_2 - x_3 &= 0 \\ -x_2 + 2x_3 - x_4 &= 0 \\ -x_3 + 2x_4 &= 1 \end{aligned}$$

THE END

Shahjalal University of Science and Technology, Sylhet

B. Sc. (Hons.) 3[rd] Year 2[nd] Semester Final Examination 2022

Department of Mathematics

Course No.: MAT327; Title: Numerical Analysis

Credit: 03; Time: 03 hours; Marks: 60.

[Answer five questions taking at least any two from each group. Numbers at right margin indicate marks.]

Group-A

1. (a) (i) Give the definition of the following errors: relative error, round-off errors, truncation error. (3)
   (ii) Derive a general error formula. (3)
   (b) Describe the method of iteration for finding a root of the equation $f(x) = 0$. Also establish the condition of convergence of this method. (6)

2. (a) Explain Newton-Raphson method to compute a real root of the equation $f(x) = 0$ and find the condition of convergence. Hence find a non-zero root of the equation $x^3 - 5x + 3 = 0$. (8)
   (b) Compute to four decimal places the root between 1 and 2 of the equation $x^3 - 2x^2 + 3x - 5 = 0$ by the method of false position. (4)

3. (a) Define interpolation. Derive the Newton's formula for forward interpolation. (6)
   (b) Derive the expression for the error in polynomial interpolation. Then find the error for Newton's forward difference interpolation formula. (6)

4. (a) Define divided differences. Derive Newton's general interpolation formula with divided differences. (6)
   (b) Using Lagrange's interpolation formula, find the form of the function $y = f(x)$ from the following table: (6)

x:	0	1	3	4
$f(x)$:	-12	0	12	24

Group-B

5. (a) What are spline functions and spline interpolation? (2)
   (b) Fit a cubic spline curve that passes through $(0, 0.0), (1, 0.5), (2, 2.0)$ and $(3, 1.5)$ with the natural end boundary conditions, $y''(0) = y''(3) = 0$. (10)

6. (a) Derive a formula for numerical evaluation of $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$ from a given data set of (x, y). (6)
   (b) A rod is rotating in a plane about one of its ends. The angle θ (in radians) at different tine t (seconds) are given below: (6)

t:	0	0.2	0.4	0.6	0.8	1.0
θ:	0.0	0.15	0.5	1.15	2.0	3.20

Find its angular velocity and angular acceleration when $t = 0.6$ seconds.

7. (a) Evaluate $\int_{-2}^{2} \frac{x}{5+2x} dx$ using the trapezoidal rule with nine ordinates. (6)
   (b) Evaluate $\int_0^1 \frac{dx}{1+x}$ using (i) Simpson's 1/3-rule and (ii) Simpson's 3/8-rule with $h = 1/6$. (6)

8. (a) Solve the following system of equations by Gauss elimination method with partial pivoting: (6)

$$5x - 2y + z = 4$$
$$7x + y - 5z = 8$$
$$3x + 7y + 4z = 10.$$

   (b) Solve the following tri-diagonal system: (6)

$$4x_1 - x_2 = 0.4$$
$$-x_1 + 4x_2 - x_3 = 0.8$$
$$--x_2 + 4x_3 - x_4 = 1.2$$
$$-x_3 + 4x_4 - x_5 = 1.6$$
$$-2x_4 + 4x_5 = 1.6$$

Shahjalal University of Science and Technology, Sylhet

Department of Mathematics

3rd Year 2nd Semester, Final Examination-2021

Course Title: Numerical Analysis, Course Code: MAT-327

Marks-60 Time-03 hours

(Answer any five questions taking at least two from each group)

Group-A

1. (a) What is meant by error in numerical analysis? Explain the following error: truncation error, round-off error, inherent error. 04

(b) Define bracketing interval. Discuss how the intermediate value theorem exhibits the bracketing interval. 03

(c) Find the root of $(x-4)^2(x+2)=0$ in the interval $(-2\cdot5,-1\cdot0)$ with a pre-specified tolerance $0\cdot1\%$ by using regular-falsi method. 05

2. (a) Discuss briefly the iteration method for solving an equation $f(x)=0$. Use this method, find a real root of the equation $x^3+x^2-1=0$ on the interval [0,1] with an accuracy of 10^{-4}. 05

(b) Obtain the Newton-Raphson method and show that this method converge quadratically. By applying Newton's method, find a root of the equation $x\,sinx+cosx=0$ correct upto 3d.p. 07

3. (a) Define interpolation and extrapolation. Derive the expression for the error in polynomial interpolation. 06

(b) The population of a town in decennial census were given in the following table. 06

Year	1951	1961	1971	1981	1991
Population (in thousand)	46	66	81	93	101

Estimate the population for the year 1985 using Newton's backward formula.

4. (a) Derive the Lagrange's interpolation formula for unequal intervals. Write down the advantage and disadvantage of Lagrangian interpolation. 06

(b) Use newton divided difference formula to derive interpolating polynomial for the data points (0,-1), (1, 1), (2, 9), (3, 29), (5,129) and hence compute the value of the point $y(4)$. 06

Group-B

5. (a) Determine the cubic spline polynomial for the following data set and hence compute the value of $f(0\cdot3)\,and\,f(2\cdot6)$. 12

x:	0	1	2	3
y:	1	-8	-30	-59

Use natural spline conditions. Also, verify the interpolation and continuity conditions of the cubic spline at $x=2$

6. (a) Derive a formula for numerical evaluation of $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$ from a given data set of (x, y). 06

(b) The table below gives the results of an observation, θ is the observed temperature in degrees celsius of a vessel of cooling water. t is the time in minutes from the begining of observation. 06

t:	1	3	5	7	9
θ:	85·3	74·5	67·0	60·5	54·3

Find the approximate rate of cooling at t=3.5.

7. (a) Solve the following system by Gauss-Jordan method: 05

$$x+y+z=9$$
$$2x-3y+4z=13$$
$$3x+4y+5z=40$$

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Numerical analysis

(b) Solve the following tri-diagonal system:

$$0 \cdot 5x_1 + 0 \cdot 25x_2 = 0 \cdot 35$$
$$0 \cdot 35x_1 + 0 \cdot 8x_2 + 0 \cdot 4x_3 = 0 \cdot 77$$
$$0 \cdot 25x_2 + x_3 + 0 \cdot 5x_4 = -0 \cdot 5$$
$$x_3 - 2 \cdot 0x_4 = -2 \cdot 25$$

8. (a) Obtain the general quadrature formula and hence deduce the Trapezoidal and Simpson's $^1/_3$ rules. 08
Evaluate
$\int_0^1 \frac{dx}{1+x}$ Correct to 3d.p. by Trapezoidal rule and Simpson's $^1/_3$ rules with $h = 0.25$

(b) A reservoir discharging through sluices at a depth h below the water surface has a surface area A for various values of h as given below: 04

$h(ft)$	10	11	12	13	14
$A(ft^2)$	950	1070	1200	1350	1530

If t denotes the time in minutes, the rate of fall of the surface is given by

$$\frac{dh}{dt} = \frac{-48}{A}\sqrt{h}$$

Estimate the time taken for the water level to fall from 14 ft to 10 ft above the sluices.

1st year 2nd Semester Examination-2019
Subject: Mathematics
Course No. MAT-12[illegible] Course Title: Numerical Analysis
Marks: [illegible], Time: [illegible] Hours

Answer any FIVE questions of the following. Marks distributions of each question are indicated at the beginning of the same

1. Marks: [illegible]
   (a) What is meant by error in numerical analysis? Explain the following error: Truncation error, round-off error and [illegible] error.
   (b) Define bracketing interval. Derive the formula for the [illegible] method to find a real root of a transcendental equation.
   (c) Find a root of the equation $x \log_{10} x$ [illegible] $4 \cdot$ [illegible] by Newton-Raphson method correct to three decimal places.

Marks: [illegible]
   (a) Derive the [illegible] form formulae to solve the tridiagonal system of linear equations:
   $$a_r x_{r-1} + b_r x_r + c_r x_{r+1} = f_r,\ a_1 = c_n = 0 \text{ for } r = 1, 2, 3, \ldots, n.$$
   (b) Apply above formulae to solve the following system of linear equations:
   $$0 \cdot 5x_1 + 0 \cdot 25x_2 = 0 \cdot 35$$
   $$0 \cdot 35x_1 + 0 \cdot 8x_2 + 0 \cdot 4x_3 = 0 \cdot 77$$
   $$0 \cdot 25x_2 + x_3 + 0 \cdot 5x_4 = -0 \cdot 50$$
   $$x_3 - 2 \cdot 0x_4 = -2 \cdot 25$$

3. Marks: 4+5+5
   (a) Discuss Gauss-Seidel and Gauss-Jacobi iteration methods to find solutions of the system of linear equations $Ax = b$ and write their advantages and disadvantages.
   (b) Use Gauss-Seidel iteration method to find the solutions, correct up to 3 decimal places of the following system of linear equations:
   $$4x_1 + x_2 + 2x_3 = 4$$
   $$x_1 + x_2 + 3x_3 = 3$$
   $$3x_1 + 5x_2 + x_3 = 7$$
   (c) Solve the following system of linear equations:
   $$5x - 2y + z = 4$$
   $$7x + y - 5z = 8$$
   $$3x + 7y + 4z = 10$$
   by [illegible] reduction method.

4. Marks: 6+8
   (a) Define interpolation and extrapolation. Derive the expression for the error in polynomial interpolation.
   (b) The following table gives the population of a town during the last six censuses. Estimate using any suitable interpolation formula, the increase in the population during the period from 1970 to 1978.

Year	1941	1951	1961	1971	1981	1991
Population (in thousand)	12	15	20	27	39	52

5. Marks: 7+7
   (a) Discuss briefly the properties of cubic spline and and defines its knots. Fit a cubic spline curve that passes through (0, 0 · 0), (1, 0 · 5), (2, 2 · 0) and (3, 1 · 5) with the natural-end boundary conditions. $S''(0) = 0$, $S''(3) = 0$
   (b) Find interpolating polynomial for the following data using Lagrange's formula

x	1	2	−4
$y = f(x)$	3	−5	4

   Hence find $f(2 \cdot 25)$

6. Marks: 7+7
   (a) Derive general quadrature formula to evaluate the integral $I = \int_a^b y(x)dx$, hence deduce the trapezoidal, Simpson's 1/3 and Simpson's 3/8 formula to find I
   (b) Compute the value of the definite integral $\int_{\frac{\pi}{2}}^{\pi} e^{\sin x} dx$ by Simpson's 3/8 rule and Weddle's rule. After finding the true value of the integral, compare the errors in both cases and comment which method is better.

7. Marks: 7+7
   (a) Define initial value problem. Derive Euler's method to solve the IVP:
   $\frac{dy}{dx} = f(x, y),\ y(x_0) = y_0$. Also explain its modification.
   (b) Using modified Euler's method, obtain the solution of the differential equation $\frac{dy}{dx} = x + \sqrt{y}$ with the initial condition $y(0) = 1$, for the range $0 \le x \le 0 \cdot 6$ in steps of $0 \cdot 2$.

8. Marks: 6+8
   (a) Use fourth order Runge-Kutta method to solve numerically the initial value problem:
   $\frac{dy}{dt} = y^2 - 100 \exp[-100(t-1)^2]$, $y(0 \cdot 8) = 4 \cdot 9491$ and find $y(0 \cdot 85)$ taking $h = 0 \cdot 01$
   (b) Find $y(0 \cdot 8)$ using Milne's predictor-corrector Method, if $y(x)$ is the solution of the differential equation $\frac{dy}{dx} = -xy^2$, $y(0) = 2$ assuming $y(0 \cdot 2) = 1 \cdot 92308$, $y(0 \cdot 4) = 1 \cdot 72414$ and $y(0 \cdot 6) = 1 \cdot 47059$.

The End

Shahjalal University of Science and Technology, Sylhet

B.Sc. 1st Year 2nd Semester Final Examination 2018

Subject: Mathematics; Course No: MAT 121

Course Title: Numerical Analysis

Marks: 70 Credits: 3.0 Time: 3 Hours

Answer any FIVE questions. Marks distribution of each question are indicated at the beginning of the same.

Question 1. Marks: (1+3+6+4=14)

a) Define algebraic and transcendental equations. Discuss with the intermediate value theorem to obtain the bracketing interval.

b) Describe the method of iteration for finding a root of the equation $f(x) = 0$. Also establish the condition of convergence of this method.

c) Find a real root, correct to three decimal places, of the equation $\sin^2 x = x^2 - 1$ lying in the interval [1.3, 1.5] by using iteration method.

Question 2. Marks: 7+7=14

a) Write down the iteration formula of Newton-Raphson method to find a root of the equation $f(x) = 0$. Then show that Newton-Raphson method converges quadratically.

b) Solve the equation $e^x - x - 2 = 0$ by Newton-Raphson method.

Question 3. Marks: 5+4+5=14

a) Define interpolation and extrapolation. Derive the Newton's formula for the backward interpolation.

b) Derive the first fourth backward difference formula and construct its backward difference table.

c) The population of a town in the decennial census was as given below

Year: x	1981	1991	2001	2011
Population: y (in thousands)	46	66	81	93

Estimate the population for the year 1985.

Question 4. Marks: 7+7=14

a) Derive the Lagrange's interpolation formula for unequal intervals.

b) Given the following data:

x	10.1	22.2	32.0	41.6	50.5
$f(x)$	0.17537	0.37784	0.52992	0.66393	0.63608

Estimate the value of $f(27.5)$ by using divided difference formula.

Question 5. Marks: 7+7=14

a) Derive general quadrature formula for equidistance ordinates to integrate $f(x)$ numerically and hence deduce Simpson's $\frac{1}{3}$ rule.

b) Evaluate the value of the integral $\int_{0.2}^{1.4}(\cos x - \log_e x + e^x)\,dx$ by (i) Simpson's $\frac{1}{3}$ rule and (ii) Weddle's rule.

Question 6. Marks: 8+6=14

a) Derive cubic spline interpolating method.

b) Fit a natural cubic spline to the following data

x	1	2	3
y	-8	-1	18

and compute $y(1.5)$ and $y'(1)$.

Question 7. Marks: 7+7=14

a) Use the decomposition method to solve the system of equations

$$x_1 + x_2 + x_3 = 1$$
$$4x_1 + 3x_2 - x_3 = 6$$
$$3x_1 + 5x_2 + 3x_3 = 4.$$

b) Use Gauss-Seidel iteration method to solve the following system of equations:

$$4x + y + 2z = 4$$
$$3x + 5y + z = 7$$
$$x + y + 3z = 3$$

up to four decimal places.

Question 8. Marks: 7+7=14

a) Derive Euler's method to solve the IVP:

$$\frac{dy}{dx} = f(x, y),\ y(x_0) = y_0.$$

Also explain its modification.

b) Solve the initial value problem:

$$\frac{du}{dt} = -2tu^2,\ u(0) = 1$$

with $h = 0.2$ on the interval $[0, 0.4]$ by using the fourth-order classical Runge-Kutta method.